

Quiz 4: Qualitative Analysis III — Mixtures & Equations

Review of Session 4. ~15 minutes. A calculator helps for Section C.

Section A: Mixtures

1. A sample fizzes a little with acid and leaves white residue that won't dissolve and is iodine-negative. One powder or a mixture? Of what?
2. You want to separate a salt + cornstarch mixture. Describe the steps.
3. In that separation, which component is in the **filtrate** and which is the **residue**?

Section B: Balancing

Balance by adjusting coefficients only:

1. $\underline{\hspace{1cm}} \text{H}_2 + \underline{\hspace{1cm}} \text{Cl}_2 \rightarrow \underline{\hspace{1cm}} \text{HCl}$
2. $\underline{\hspace{1cm}} \text{Al} + \underline{\hspace{1cm}} \text{O}_2 \rightarrow \underline{\hspace{1cm}} \text{Al}_2\text{O}_3$
3. $\underline{\hspace{1cm}} \text{CH}_4 + \underline{\hspace{1cm}} \text{O}_2 \rightarrow \underline{\hspace{1cm}} \text{CO}_2 + \underline{\hspace{1cm}} \text{H}_2\text{O}$
4. $\underline{\hspace{1cm}} \text{NaHCO}_3 + \underline{\hspace{1cm}} \text{HCl} \rightarrow \underline{\hspace{1cm}} \text{NaCl} + \underline{\hspace{1cm}} \text{H}_2\text{O} + \underline{\hspace{1cm}} \text{CO}_2$
5. Why can't you balance an equation by changing a subscript?

Section C: Moles

1. What is the molar mass of CO_2 ? (C = 12, O = 16)
 2. What is the molar mass of NaHCO_3 ? (Na = 23, H = 1, C = 12, O = 16)
 3. What is the formula: moles = $\underline{\hspace{2cm}} \div \underline{\hspace{2cm}}$?
 4. In a reaction, how do you find the limiting reactant?
 5. You have 0.30 mol of reactant A (coefficient 2) and 0.20 mol of reactant B (coefficient 1). Which is limiting?
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